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$$\int \frac{dx}{4x^2 + 12x + 13} = \int \frac{dx}{(2x + 3)^2 + 2^2}$$

$$t = 2x + 3$$

$$dt = 2dx$$

$$dx = \frac{dt}{2}$$

$$\int \frac{\frac{dt}{2}}{t^2 + 2^2} = \frac{1}{2} \cdot \int \frac{dt}{t^2 + 2^2} = \frac{1}{2} \cdot \frac{1}{2} \cdot \text{Bgtan} \left(\frac{t}{2} \right) + c = \frac{1}{4} \cdot \text{Bgtan} \left(\frac{2x + 3}{2} \right) + c$$

References